Week 3: Sentiment & News Alphas, and the Road to Gold

- Dates: 2026-10-20 → 2026-10-24 (midterm week)
- Scope: BRAIN Tutorial Chapter 5 (complete) and Chapter 6 (in progress: reach 10,000 Challenge points)
- Simulations: 25, 3 submitted this week (6 in total)
- Common settings: USA · D1 · TOP3000 · Truncation 0.08 · NaN handling On · Test period 0 (full in-sample period)

Summary

Chapter 5 introduced **social-media sentiment** and **news** data. News is stored as *vector* data (several items per stock per day), so it has to be collapsed with `vec_avg` before normal operators can use it. Chapter 6 has a single task: keep submitting until the Challenge score reaches 10,000 (Gold). That turned out to be the hard part. **Variations of an alpha I had already submitted were blocked by the self-correlation check, even when they performed better.** Every new submission needed a genuinely new idea. New ideas were usually weak on their own, but blending two or three of them with different characteristics got them over the bar.

The score went from 3,876 to 7,690, with one more submission pending and the next candidate ready.

Roadmap

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Chapter 5 – Sentiment & News
  social buzz vs volume (ts_regression)   Sharpe ±1.06, direction flips in 2022
  news "long-favoured" → momentum, else 5-day reversal
                                          Sharpe 1.91  --> SUBMITTED (Decay 8)
Chapter 6 – Expand ideas, reach Gold
  universe / neutralization / signed_power / trade_when on the news alpha
                                          up to Sharpe 2.03, but self-correlation 0.94 → blocked
  analyst-revision (reversed) + financial health (FFO / debt)
                                          0.90 + 0.67 → 1.34  --> SUBMITTED
  leverage alpha from week 2, re-checked on the full period
                                          Sharpe 1.55  --> SUBMITTED
  option IV gap + liability fair value + leverage
                                          Sharpe 1.90, Fitness 1.81 --> next candidate
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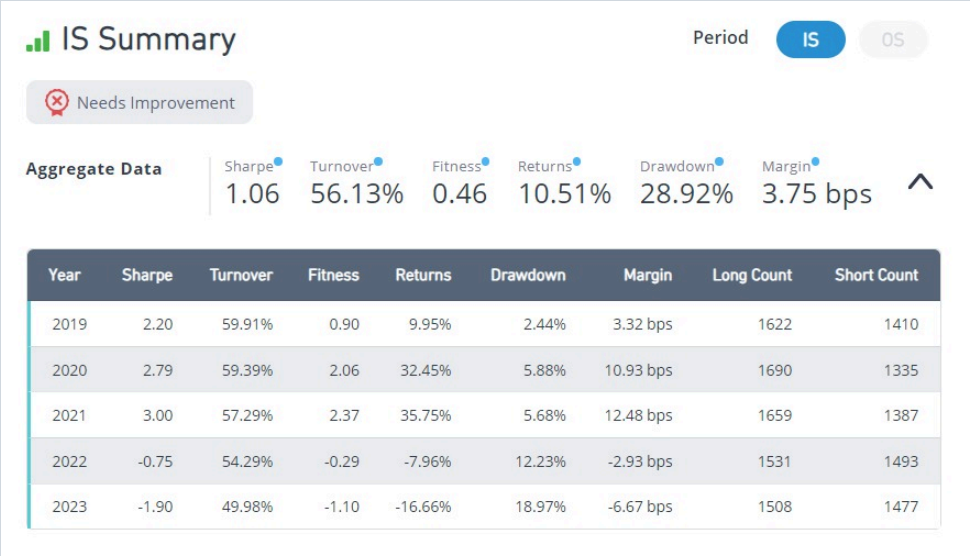
1. Sentiment: social buzz versus trading volume

Idea: if a stock is discussed online far more than its trading volume would suggest, attention may be overheated.

`ts_regression(y, x, d)` fits a line of `y` on `x` over the last `d` days and returns the residual, here "how much more buzz than volume explains".

Expression	Sharpe	2019	2020	2021	2022	2023
`ts_regression(scl12_buzz, volume, 250)`	-1.06	-2.20	-2.79	-3.00	+0.75	+1.90
`-ts_regression(scl12_buzz, volume, 250)`	1.06	**+2.20**	**+2.79**	**+3.00**	-0.75	-1.90

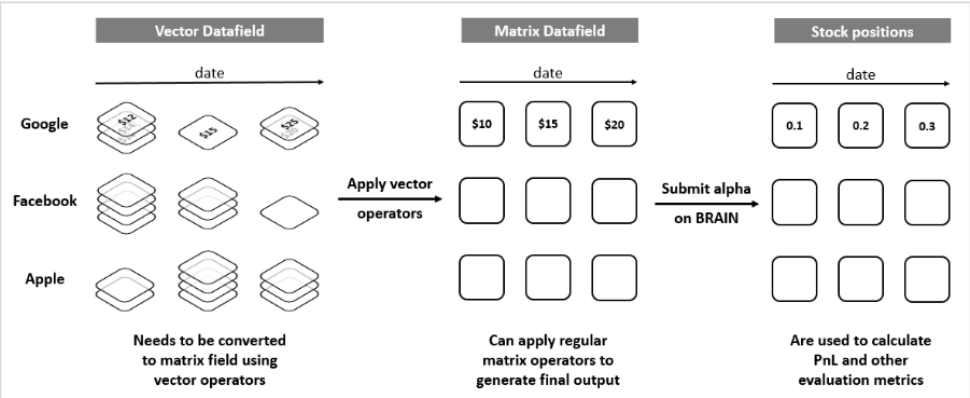
Flipped, the signal was very strong through 2021 (excess attention → price falls) and then **reversed direction from 2022**. The aggregate Sharpe of 1.06 hides a regime change; only the yearly table reveals it. I did not use it further.



Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.20	59.91%	0.90	9.95%	2.44%	3.32 bps	1622	1410
2020	2.79	59.39%	2.06	32.45%	5.88%	10.93 bps	1690	1335
2021	3.00	57.29%	2.37	35.75%	5.68%	12.48 bps	1659	1387
2022	-0.75	54.29%	-0.29	-7.96%	12.23%	-2.93 bps	1531	1493
2023	-1.90	49.98%	-1.10	-16.66%	18.97%	-6.67 bps	1508	1477

2. News: vector data and a conditional alpha

News data covers earnings reports, analyst calls, product approvals and legal events, mostly published before or after the main session. Because one stock can have several news items per day, the fields are **vectors**. `vec_avg` turns each day's items into a single number.



`nws12_afterhsz_sl` tells whether going long or short was more profitable after each news release.

Idea: stocks whose recent news favoured longs show momentum; the rest tend to revert. One conditional expression combines both:

- 60-day average of `vec_avg(nws12_afterhsz_sl)`, ranked
- top half → constant long (momentum); bottom half → 5-day price reversal

Version	Sharpe	Turnover	Fitness	Margin
Decay 4 (tutorial step)	1.95	22.73%	1.12	6.58 bps
Decay 8	1.91	**16.08%**	**1.25**	**8.60 bps**

Raising Decay cut turnover by a third with almost no loss of Sharpe, the same lesson as week 1.

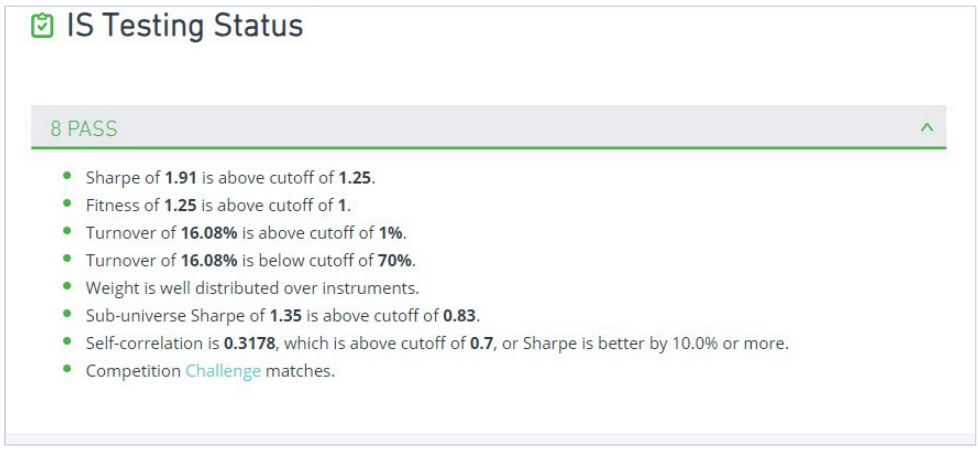
Submitted: news momentum / reversal ✓

Item	Detail
Operators	`vec_avg`, `ts_mean`, `rank`, `ts_delta`, conditional `? :`
Data fields	`nws12_afterhsz_sl`, `close`
Settings	TOP3000 · Subindustry · Decay 8
Result (2019–2023)	**Sharpe 1.91 · Fitness 1.25 · Turnover 16.08%** · Returns 6.92% · Drawdown 3.02%
Checks	8 passed. Sub-universe 1.35, self-correlation 0.32

> For submitted alphas I share the idea, operators and data fields, but not the exact expression.



Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.70	16.24%	1.73	6.67%	2.02%	8.22 bps	1565	1450
2020	4.16	16.28%	4.09	15.72%	1.14%	19.32 bps	1577	1432
2021	1.59	16.00%	1.03	6.66%	2.48%	8.33 bps	1639	1423
2022	0.32	16.00%	0.09	1.39%	3.02%	1.73 bps	1634	1426
2023	1.66	15.90%	0.92	4.87%	1.17%	6.13 bps	1654	1421





3. Chapter 6: expanding an idea, and why it was not enough

The tutorial lists four ways to get more alphas from one idea. I applied each to the news alpha:

Method	Change	Sharpe	Fitness	Outcome
(baseline)	submitted alpha	1.91	1.25	submitted
Universe	TOP1000	1.44	0.83	fails — signal lives in smaller stocks
Neutralization	Industry	1.73	1.17	passes thresholds
Position distribution	reversal leg → `signed_power(·, 2)`	**2.03**	**1.35**	best
Synergy	`trade_when(volume > adv20, ·, -1)`	1.54	0.90	fails
Size neutralization	`group_neutralize` by market-cap deciles (`regression_neut` is not available at my level)	1.94	1.25	**self-correlation 0.94 → blocked**

The catch: BRAIN rejects a new alpha whose correlation with one of my submitted alphas exceeds 0.7, unless its Sharpe is at least 10% higher. Every variation above was essentially the same alpha. Better metrics did not help.



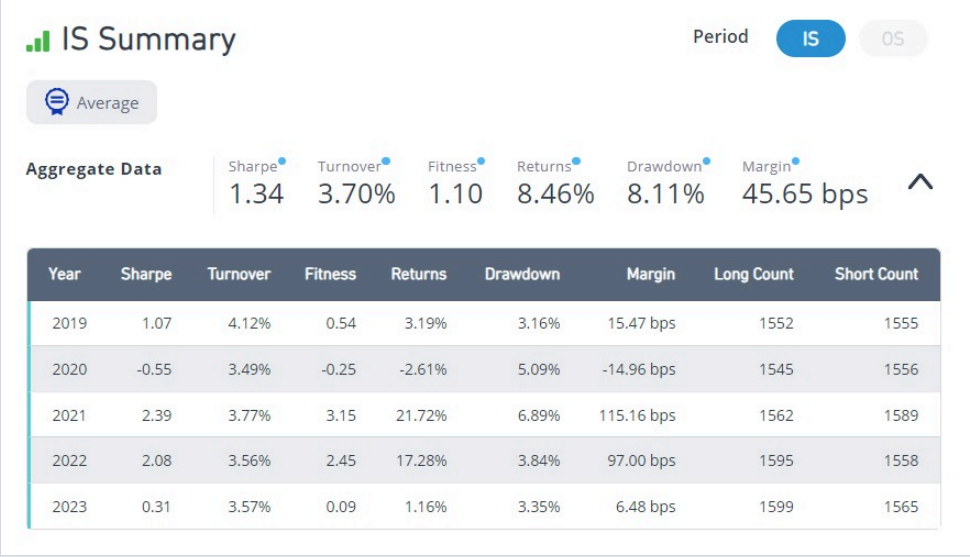
4. Finding new ideas

When tuning stopped helping, I searched the web (analyst estimate revisions are an unused data category; a common recipe is signal → scale by volatility → neutralize). I also used BRAIN's built-in **Example** alphas. Note: loading an example also resets the simulation settings.

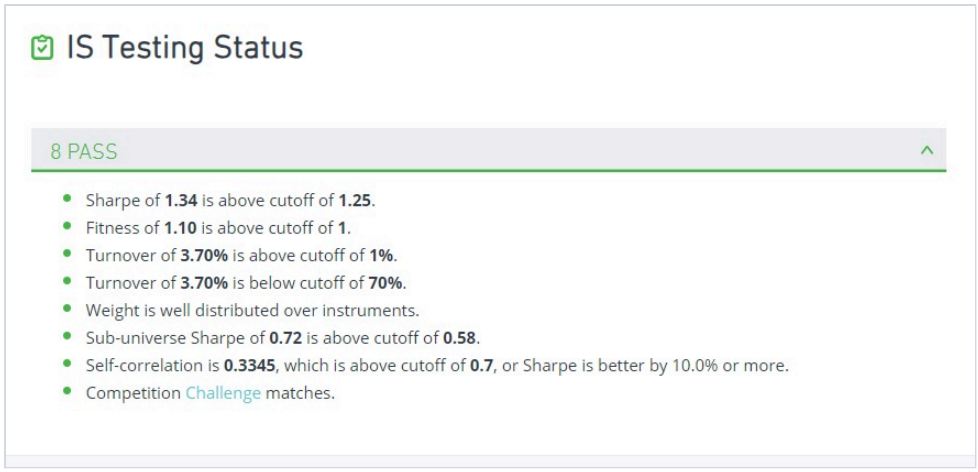
#	Idea	Sharpe	Outcome
12	20-day change in consensus EPS ÷ price	-0.13	no signal
13	Analyst dispersion (std ÷ mean of estimates)	-0.12	collapsed in 2020
14	`-rank(ebit/capex)` (BRAIN example)	-0.20	fails
15	Funds from operations ÷ long-term debt, 1-year time-series rank	0.67	weak but positive
16→17	Analyst-revision momentum score, then reversed	-0.90 → 0.90	over-reaction to revisions reverts
18	17 + 15, 50:50	1.32	8 checks passed
19	18 with Industry neutralization	**1.34**	**submitted**

Submitted: analyst-revision reversal + financial health ✓

Item	Detail
Idea	Fade stocks whose analyst estimates were revised up most (over-reaction) and buy companies whose cash flow relative to debt is improving versus their own past
Data fields	`composite_factor_score_derivative`, `fnd6_fopo`, `debt_lt`
Settings	TOP3000 · Industry · Decay 8
Result	**Sharpe 1.34 · Fitness 1.10 · Turnover 3.70%** · Margin 45.65 bps
Checks	8 passed. Self-correlation 0.33



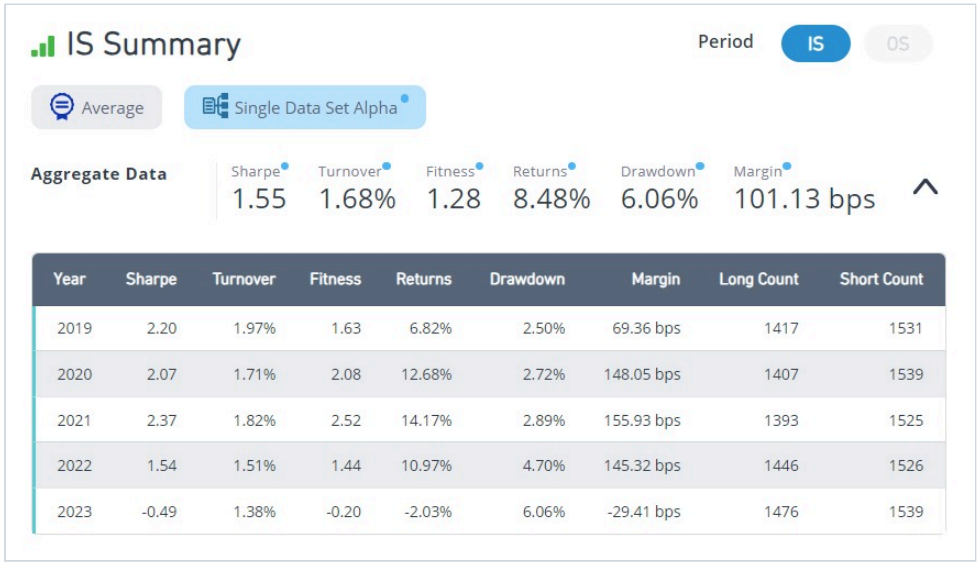
Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	1.07	4.12%	0.54	3.19%	3.16%	15.47 bps	1552	1555
2020	-0.55	3.49%	-0.25	-2.61%	5.09%	-14.96 bps	1545	1556
2021	2.39	3.77%	3.15	21.72%	6.89%	115.16 bps	1562	1589
2022	2.08	3.56%	2.45	17.28%	3.84%	97.00 bps	1595	1558
2023	0.31	3.57%	0.09	1.16%	3.35%	6.48 bps	1599	1565





Submitted: leverage, revisited ✓

The week 2 alpha `rank(-equity/assets)` (more leveraged companies within a subindustry outperform) was never submitted. On the full 2019–2023 period it scored **Sharpe 1.55, Fitness 1.28, Turnover 1.7%** and passed all eight checks (self-correlation 0.66).



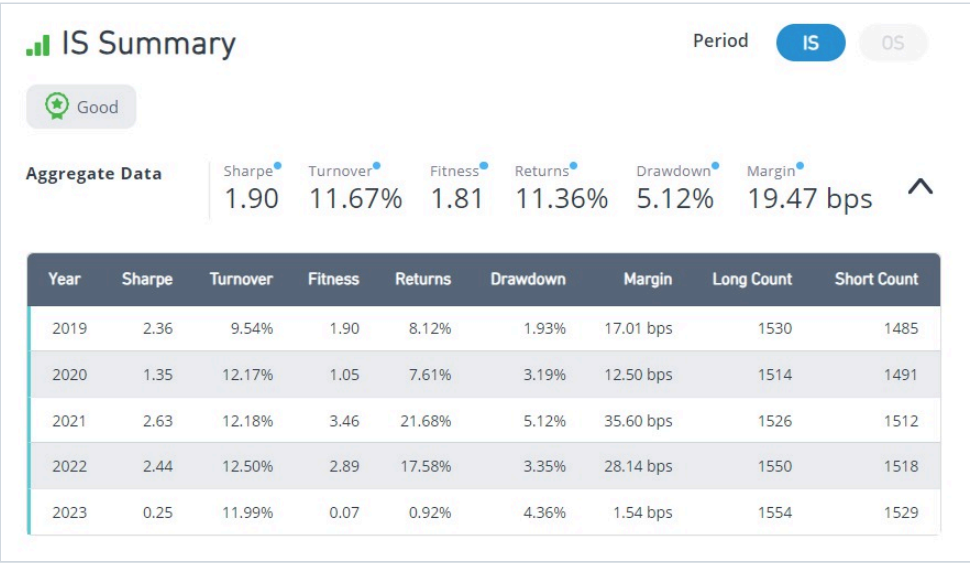
Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.20	1.97%	1.63	6.82%	2.50%	69.36 bps	1417	1531
2020	2.07	1.71%	2.08	12.68%	2.72%	148.05 bps	1407	1539
2021	2.37	1.82%	2.52	14.17%	2.89%	155.93 bps	1393	1525
2022	1.54	1.51%	1.44	10.97%	4.70%	145.32 bps	1446	1526
2023	-0.49	1.38%	-0.20	-2.03%	6.06%	-29.41 bps	1476	1539



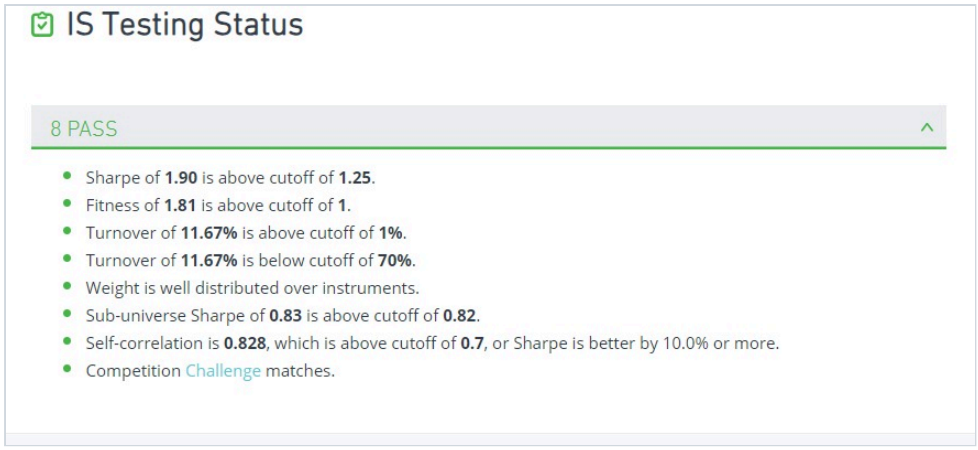
5. Next candidate: three signals blended

#	Attempt	Sharpe	Outcome
21	Low-volatility anomaly	0.13	crashed in 2020
22	Increase in fair value of liabilities → short (BRAIN example, shorter window)	0.91	only ~500 stocks covered
23	22 + option call–put IV gap	1.67	sub-universe 0.47 fails
24	23 tilted toward large caps (× `rank(cap)`)	1.09	fixes sub-universe, kills the signal
25	23 + leverage signal (strong in large caps)	**1.90**	**8 checks passed, Fitness 1.81**

#25 has a self-correlation of 0.83, but passes because its Sharpe is more than 10% above the correlated alpha. Its sub-universe Sharpe (0.83 vs a 0.82 cutoff) is right at the edge. It will be submitted next, since the daily point cap was already used.



Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.36	9.54%	1.90	8.12%	1.93%	17.01 bps	1530	1485
2020	1.35	12.17%	1.05	7.61%	3.19%	12.50 bps	1514	1491
2021	2.63	12.18%	3.46	21.68%	5.12%	35.60 bps	1526	1512
2022	2.44	12.50%	2.89	17.58%	3.35%	28.14 bps	1550	1518
2023	0.25	11.99%	0.07	0.92%	4.36%	1.54 bps	1554	1529



Submitted alphas so far

Date	Week · chapter	Ingredients	Idea	Sharpe	Fitness
10/11	1 · Ch2	price + fundamentals	Short-term reversal + profitability	2.46	1.91
10/17	2 · Ch3	fundamentals + price	Earnings-yield time-series rank (TOP200)	1.64	1.76
10/17	2 · Ch4	options + fundamentals	Call–put IV gap + earnings yield	2.49	1.69
10/20	3 · Ch5	**news** + price	News-conditioned momentum / reversal	1.91	1.25
10/21	3 · Ch6	**analyst** + fundamentals	Revision reversal + financial health	1.34	1.10
10/24	3 · Ch6	fundamentals	Leverage	1.55	1.28

What I learned this week

1. **Vector data needs a `vec_` operator first.** Once collapsed, one conditional expression can switch between momentum and reversal.
2. **Read the yearly table, not just the total.** The sentiment signal flipped sign in 2022, and its total Sharpe hid that.
3. **A better alpha is not necessarily a submittable one.** Variations of a submitted idea hit self-correlation 0.94. The exception is a Sharpe at least 10% higher than the correlated alpha.
4. **New ideas start weak; blending makes them work.** Twice this week, two or three signals that each failed on their own (0.67–0.91) passed together (1.34, 1.90).
5. **Fix sub-universe failures with a large-cap-friendly signal, not a size tilt.** Multiplying by market cap fixed the check but destroyed the signal.

Next week

- Submit the three-signal blend and reach Gold (10,000 points)
- Look for ingredients not yet used: volume patterns, other universes (TOP1000 / TOP500)

All experiments

#	Idea	Neut. / Decay	Sharpe	Turnover	Fitness	Note
1	Buzz vs volume regression (250d)	Sub / 4	-1.06	56%	-0.46	
2	#1 reversed	Sub / 4	1.06	56%	0.46	flips in 2022
3	News-conditioned momentum / reversal	Sub / 4	1.95	23%	1.12	8 passed
4	#3, Decay 8	Sub / 8	1.91	16%	1.25	submitted
5	#4 in TOP1000	Sub / 8	1.44	17%	0.83	
6	#4, Industry	Ind / 8	1.73	15%	1.17	
7	#4 with `signed_power` reversal	Sub / 8	2.03	14%	1.35	
8	#7 with `trade_when`	Sub / 8	1.54	11%	0.90	
9	#7, market-cap group neutralized	Sub / 8	1.94	14%	1.25	self-corr 0.94
10	#2 with 20-day window	Sub / 8	0.58	42%	0.21	
11	Implied ÷ historical volatility, reversed	Sub / 8	-0.79	4%	-0.42	
12	EPS estimate change ÷ price	Sub / 8	-0.13	12%	-0.03	
13	Analyst dispersion	Sub / 8	-0.12	5%	-0.03	
14	EBIT ÷ capex (example)	default	-0.20	1%	-0.09	settings reset
15	FFO ÷ LT debt, time-series rank	Sub / 8	0.67	4%	0.23	
16	Analyst-revision score	Sub / 8	-0.90	2%	-0.63	
17	#16 reversed	Sub / 8	0.90	2%	0.63	
18	#17 + #15	Sub / 8	1.32	4%	1.01	8 passed
19	#18, Industry	Ind / 8	1.34	4%	1.10	submitted
20	Leverage (week 2 alpha)	Sub / 4	1.55	2%	1.28	submitted
21	Low volatility	Sub / 4	0.13	2%	0.06	
22	Liability fair value, reversed	Sub / 4	0.91	7%	0.74	
23	#22 + IV gap	Sub / 8	1.67	18%	1.06	sub-universe fails
24	#23 × rank(cap)	Sub / 8	1.09	14%	0.62	
25	#23 + leverage	Sub / 8	1.90	12%	1.81	next candidate